

Figure 2: ETSI NFV architecture reference model

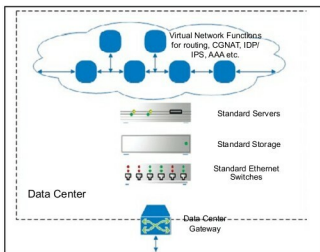


Figure 3: Service provider networks in the future (NFV based)

is created, or steered to other VM instances, when a VM is removed. SDN enables rapid traffic steering by helping the SDN controllers to modify the actions to be taken by SDN data plane elements for individual traffic flows.

The need for rich services

With the cost of bandwidth being lowered, service providers aim to improve revenues by providing a rich set of value added services. In current networking equipment, services are typically handled in a separate 'service plane', kept in the same chassis as the data and control plane components. With increasing line-card capacity, the service plane complex becomes more and more overwhelmed. To remedy this, services are provided via dedicated high performance equipment, as shown in Figure 1. These service equipment are chained together to provide a set of services, and this is called 'service chaining'. Today, it is static in nature and depends on how the devices are connected.

However, once services are moved out of the chassis, it also creates the possibility to run the same services as virtual instances running in server platforms. With SDN, we can create service chains dynamically. This provides a lot of flexibility and operational convenience to service providers. Thus, 'service chaining' is a primary driver for service providers to consider NFV, while retaining the investments already made by them in high-capacity routers and switches.

Other factors

The service providers are looking for standards based solutions so that they can get the best-of-breed equipment and avoid vendor lock-in. As the network becomes software-centric rather than hardware-centric, they are considering open source solutions as well.

The emergence of Ethernet, once considered a LAN technology, into the WAN arena also helps. Ethernet speeds have been increasing with constant innovations in the physical (optical) layers. Today we talk of servers with 40G ports and

even 100G ports. This means that servers running virtualised network appliances no longer need other equipment to terminate protocols such as SONET and ATM. They can directly terminate network traffic and provide service.

ETSI architecture for NFV

In 2012, the European Telecommunications Standards Institute (ETSI) created an industry standards group (ISG), supported by some service providers, to define requirements, use-cases, architectural references and specifications in the area of NFV.

Since then, ETSI has published various documents, defining a reference architectural model, shown in Figure 2, and various recommendations to enable NFV adoption and interoperability.

The NFVI block shown at the bottom of Figure 2 comprises the hardware, i.e., physical servers, storage devices and physical network comprising networking hardware and interconnects. Above this lies the hypervisor layer, which provides virtualised compute, storage and networks for running applications. Various Layer 2 tunnelling technologies (e.g., VXLAN, NVGRE) help create a virtual network overlay over the physical network.

The virtual network functions (VNFs), managed by their respective element management systems (EMS), run above this virtualised infrastructure. The VNFs provide the functionality that is currently provided by physical network hardware.

The NFV management and orchestration layer (MANO) comprises the orchestrator, virtual machine manager and virtualised infrastructure manager. The orchestrator works with the OSS/BSS system to determine load and network services for which VNFs need to be created or deleted. It works with the virtual machine manager to create/remove VNFs and with a virtual infrastructure manager to assign hardware, storage and network resources to the VNFs. It also steers traffic to the VNFs and creates service chains by connecting multiple service VNFs.

Figure 3 illustrates what a future service provider network based on cloud technologies would look like.